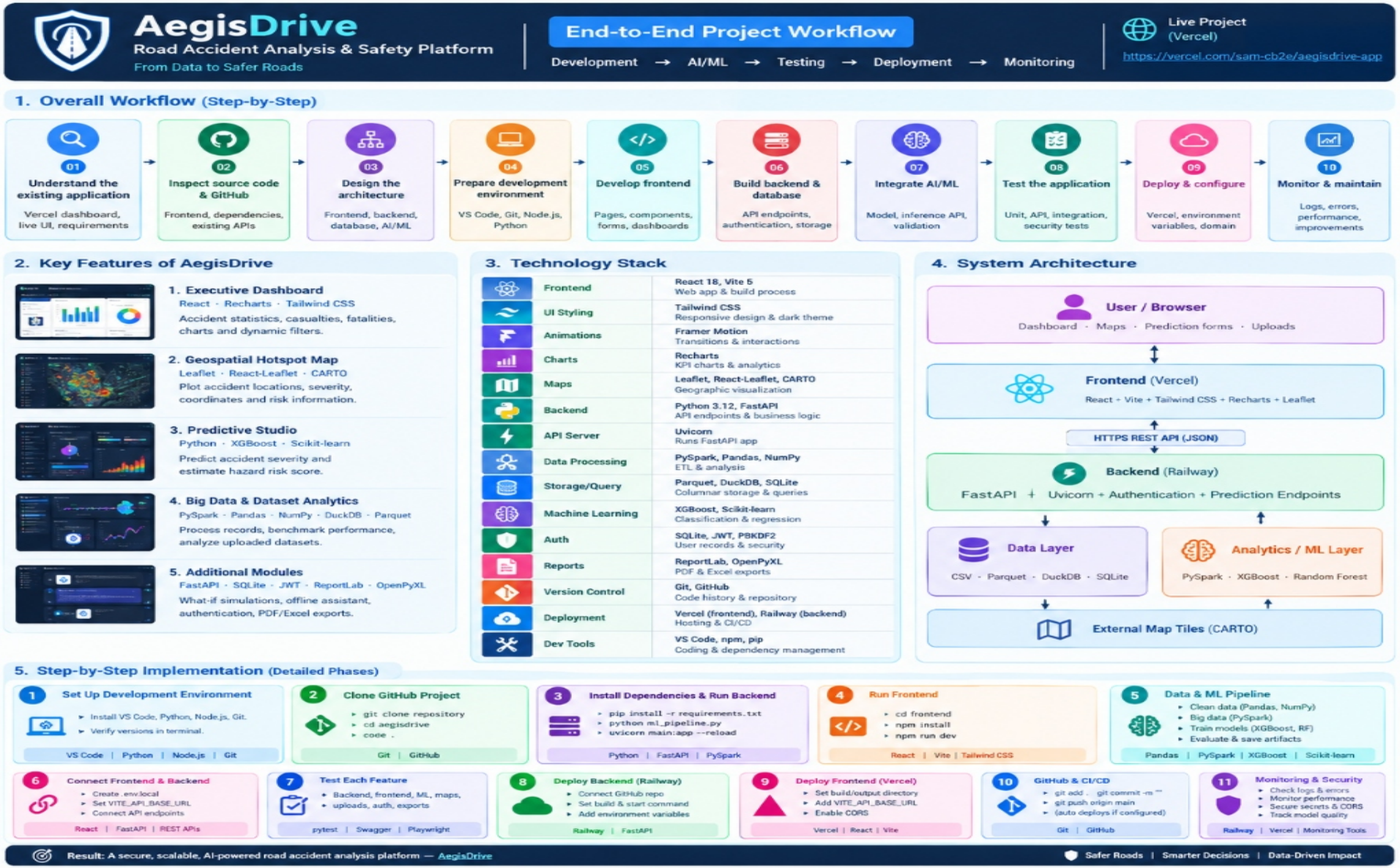
AegisDrive — Road Accident Analysis & Safety Platform

End-to-End Project Workflow & Architectural Implementation Blueprint | From Data to Safer Roads

Frontend (Live): https://aegisdrive-app.vercel.app | **Backend API (Live):** https://aegisdrive-production.up.railway.app | **GitHub:** https://github.com/Samrudhi-J07/aegisdrive



1. Detailed Step-by-Step Implementation Workflow (Phases 1 — 11)

AegisDrive was developed and deployed following a rigorous 11-stage full-stack engineering lifecycle, spanning dataset engineering, distributed analytics, machine learning pipeline training, reactive frontend UI, and multi-cloud automated CI/CD deployment.

Phase & Scope	Core Objectives & Actions Taken	Tools & Technologies	Output / Artifacts
Phase 1: Environment Setup	Configure local workstation, ensure Python 3.12, Node.js 18+, Git, and virtual environments.	VS Code, Windows PowerShell, Git 2.55	Clean isolated dev environment
Phase 2: Repository Architecture	Initialize Git repo, configure root .gitignore (protecting models & secrets), create directory hierarchy.	Git, GitHub CLI, PowerShell	Monorepo structure (backend/ + frontend/)
Phase 3: Backend & API Core	Develop FastAPI ASGI application, CORS middleware, REST routers for analytics, upload, and chat.	FastAPI, Uvicorn, Pydantic, Python 3.12	Interactive OpenAPI docs at /docs
Phase 4: Big Data & Parquet Cache	Clean 20,000+ accident records, compute derived risk scores, convert CSV to columnar Parquet for sub-second queries.	Pandas, DuckDB, PyArrow, Parquet	Optimized processed_accidents.parquet
Phase 5: Machine Learning Suite	Train XGBoost multi-class classifier (Severity: Minor, Major, Fatal) and Random Forest Risk Regressor (0.00-1.00).	XGBoost, Scikit-Learn, PyTorch, Joblib	severity_model.pkl, risk_model.pkl
Phase 6: Frontend Development	Build single page application with Cyberpunk Midnight Blue theme, executive KPI cards, charts, and Leaflet map.	React 18, Vite 5, TailwindCSS, Framer Motion	Modular UI components with micro-interactions
Phase 7: Geospatial Hotspot Map	Render 20,000+ coordinates across India with severity color codes (Red/Orange/Green) on CARTO dark tiles.	Leaflet, React-Leaflet, CARTO Basemaps API	High-contrast watermark-free spatial map
Phase 8: What-If & Dataset Uploader	Develop interactive safety simulator with speed/weather sliders and universal CSV/Excel/PDF upload auto-EDA.	ReportLab, OpenPyXL, FormData	Dynamic policy advice & instant PDF exports
Phase 9: Offline AI & Security	Build Aegis AI assistant with 10 NLP intent categories and JWT authentication with PBKDF2 password hashing.	Regex NLP, SQLite, HMAC SHA-256	Floating conversational widget & User Auth
Phase 10: Backend Cloud Deploy	Containerize backend with Python 3.12-slim Dockerfile, deploy to Railway, expose public port 8080.	Docker, Railway, Uvicorn	Live API: aegisdrive-production.up.railway.app
Phase 11: Frontend Cloud Deploy	Deploy Vite SPA to Vercel Edge network, link VITE_API_BASE_URL and CARTO credentials, test live integration.	Vercel Edge CDN, Git Push CI/CD	Live Web App: aegisdrive-app.vercel.app

2. Technology Stack & Architectural Specifications

Category	Technology / Tool	Version	Role & Capabilities
Frontend Framework	React + Vite	18.3 / 5.4	Single Page Application core, ultra-fast HMR and bundling
Styling & UX	TailwindCSS + Framer Motion	3.4 / 11.0	Midnight Blue glassmorphic theme, fluid spring animations
Data Visualization	Recharts + Leaflet	2.12 / 1.9	Declarative SVG charts (Area, Bar, Pie) and spatial heatmaps
Backend API	FastAPI + Uvicorn	0.110 / 0.30	Asynchronous REST gateway, OpenAPI schemas, high concurrency
Machine Learning	XGBoost + Scikit-Learn	2.0 / 1.4	Severity multi-class classification and continuous hazard scoring
Big Data Engine	Apache PySpark + DuckDB	3.5 / 0.10	Distributed ETL, windowed analytics, sub-second aggregation
Authentication	PBKDF2 SHA-256 + JWT	StdLib	Zero-dependency secure authentication & role management
Report Generators	ReportLab + OpenPyXL	4.1 / 3.1	Automated multi-page PDF export and Excel workbook generation
Cloud & Hosting	Vercel + Railway + Docker	Latest	Global Edge CDN frontend and containerized Python backend

3. Core Modules & Production Capabilities

Module	Key Features & Capabilities	API Endpoint / Data Flow
Executive Dashboard	Macro accident KPIs (20,000+ crashes, 34,529 casualties), hourly frequency curves, dynamic filter toolbar (Search, State, Severity, Weather), one-click PDF & Excel exports.	GET /api/analytics/summary GET /api/analytics/charts GET /api/analytics/export-pdf
Geospatial Hotspot Map	Visualizes 1,200+ coordinate clusters with severity color tags (Fatal=Red, Major=Orange, Minor=Green), accident causality popups, state filters, and CARTO dark raster basemaps.	GET /api/analytics/map CARTO Basemaps API integration
Predictive Studio	Real-time inference calculating multi-class severity probabilities and risk indices from environmental features (visibility, lanes, rush hour, weather, crash cause).	POST /api/predict/severity XGBoost & Random Forest pipeline
What-If Simulator	Scenario playground with dynamic sliders for traffic speed multipliers, adverse weather, and peak hours. Generates actionable road safety policies automatically.	POST /api/simulation/what-if Dynamic multiplier calculation
Upload & Analyze	Universal data ingestion (.csv, .xlsx, .json, .pdf) with automated data health score (0-100), missing value analysis, 10+ generated charts, preview table, and PDF download.	POST /api/upload/analyze POST /api/upload/export-pdf
Aegis AI Chatbot	100% offline conversational assistant with 10 NLP intent categories (STATS, TOP_N, TIME, CAUSE, PREDICTION, etc.), in-chat progress bars, and deep-link navigation buttons.	POST /api/chat/message In-memory vectorized aggregation
User Authentication	Secure user accounts with PBKDF2 password hashing, JWT token authorization, SQLite persistence, role profiles (Analyst, Engineer, Admin), and persistent sessions.	POST /api/auth/signup POST /api/auth/login